ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #1

WinCraft v0.1.0

Pure-Rust Windows Host, RAM Weeder, Restart Manager File-Lock Inspector, High-Speed FileOps & NSSM-Safe SysAdmin Studio



26	5 → 1	0 MB	6 Formats
MCP TOOLS & VERBS	REPOS CONSOLIDATED	IDLE DAEMON TAX	CARDS/LIST/JSON/CSV/MD/TXT

Repository: `D:\Dev\repos\wincraft` (`https://github.com/sandraschi/wincraft`) ·
Architecture: Storytold ArtCraft 3-Tier Standard · **Compiled via:** `BookCraft v0.5.0`

1. Executive Summary & Fleet Consolidation

Prior to **WinCraft**, Windows host inspection, file-lock debugging, duplicate file reclamation, and service administration were split across five separate Python MCP servers plus a prototype Python scaffold. Running multiple Python virtual environments and background webservers merely to inspect processes or check which process locked a file consumed hundreds of megabytes of RAM.

> **One Native Binary, Three Surfaces:** WinCraft compiles to a single pure-Rust workspace (`wincraft-core`, `wincraft-mcp`, `wincraft-app`) with zero runtime dependencies, zero network port claims, and strict compile-time denial of `unwrap`, `expect`, `panic`, `todo`, and `unimplemented`.

Legacy Python Repo / Scaffold	WinCraft Subsystem Replacement	Architectural Upgrade
`windows-operations-mcp`	`sysadmin` + `proc_ports`	Native NSSM-aware `sc.exe` controller, `HKCU` `.reg` auto-backups, Startup autorun & Event Log tailing.
`system-admin-mcp`	`proc_ports::weeder_scan` / `weeder_reap`	Instant RAM Weeder detecting orphaned `<crate>-<16hex>.exe` cargo test binaries and zombie workers with a hard-coded protected system allowlist.
`sysinternals-mcp`	`locks` (Win32 `rstrtmgr.dll` FFI)	Direct Win32 Restart Manager FFI (`RmStartSession` / `RmGetList`) answers "*Who locks this file?*" in microseconds without kernel drivers.
`filesystem-mcp` & `filecraft`	`fileops` + `hash`	3-stage streaming SHA-256 duplicate finder (Size → 4 KB Prefix →Full FIPS 180-4 SHA-256), disk usage breakdown, and `.bak` write/undo.
`winrar-mcp`	`fileops::fs_archive_*`	Pure-Rust deflated `.zip` creation & inspection with automatic `Rar.exe` / `7z.exe` toolchain probing in `win_doctor`.

2. Flagship Subsystems

🔒 Native Win32 Restart Manager Lock Inspector

When Windows refuses to delete or overwrite a file or directory, `win_lock_inspect` links directly against `rstrtmgr.dll` (`RmStartSession`, `RmRegisterResources`, `RmGetList`) to report the exact locking PID, application name, and service short name—with zero EULA prompts or kernel drivers.

🧬 3-Stage Streaming Duplicate Detector

`win_fs_dups` scans directory trees at native speed by (1) grouping files by exact byte size, (2) hashing only the first 4 KB of size-colliding files, and (3) streaming 64 KB chunks through a zero-dependency FIPS 180-4 SHA-256 engine only for prefix matches.

🧹 RAM Weeder & Protected Process Guard

`win_weeder_scan` identifies forgotten Rust test executables, orphaned headless browsers, and runaway workers. `win_weeder_reap` requires `confirm=true` and enforces an immutable allowlist protecting `csrss.exe`, `lsass.exe`, `svchost.exe`, `explorer.exe`, `nssm.exe`, `pwsh.exe`, and IDE hosts.

🔌 Fleet-Correlated Port Mapper

`win_port_map` maps listening TCP/UDP sockets to owning PIDs and automatically cross-references `mcp-central-docs/operations/WEBAPP_PORTS.md` so fleet services are badged by name in both the CLI and L3 desktop app.

3. Universal Cards / List / JSON Engine & Multi-Format Export

Every dataset in WinCraft (`processes`, `services`, `ports`, `disks`, `weeder` candidates, `fileops` items/duplicates/grep matches, `locks`, `startup`, `tasks`, `registry`, and `tools`) flows through `wincraft_core::export`:

- **Switchable Views (Default: Cards):** Toggle instantaneously between **🃏 Cards (Default)** (with visual capacity/memory bars and one-click action buttons), **☰ List / Table**, and `{ } JSON`.
- **Live Substring Filtering & Numeric-Aware Sorting:** Filter across all columns in real time and sort any column ascending or descending with automatic numeric detection (PID, MB, GB, %, Port).
- **Full Pagination:** Choose `10`, `25`, `50`, or `100` items per page with `First`, `Prev`, `Page X / Y`, `Next`, and `Last` navigation.
- **One-Click & CLI Export:** Export any filtered, sorted view to `CSV`, `MD` **(Markdown Table)**, `TXT`, or `JSON`.

```
# Default Cards view (paginated, 5 items per page)
wincraft proc-list --page-size 5

# List/Table view sorted by Display Name (col 1)
wincraft svc-list --format list --sort-col 1 --page-size 15

# Markdown table output + simultaneous CSV export
wincraft disk-list --format md --export D:\Dev\repos\wincraft\exports\disks.csv
```

4. Complete 26-Tool Reference Matrix

All 26 tools are exposed over stdio MCP (accepting both `win_*` and dotted `win.*` names) and as CLI subcommands.

MCP Tool	CLI Verb	Domain	Mode	Confirm	Capability
`win_status`	`status`	system	Read-Only	No	Host overview: logical volumes, process count, RAM weeder summary, PATH health.
`win_doctor`	`doctor`	system	Read-Only	No	Verify Restart Manager FFI, `sc.exe`, `reg.exe`, `wevtutil.exe`, `Rar.exe`, `7z.exe`, Sysinternals.
`win_tools`	`tools`	system	Read-Only	No	List all 26 MCP tools and their safety metadata.
`win_disk_list`	`disk-list`	system	Read-Only	No	Enumerate logical volumes (`A:..Z:`) via Win32 `GetDiskFreeSpaceExW`.
`win_env_audit`	`env-audit`	system	Read-Only	No	Audit `PATH` for dead/missing directories and duplicate entries.
`win_proc_list`	`proc-list`	processes	Read-Only	No	List running Windows processes sorted by working-set memory with protected status.
`win_port_map`	`port-map`	processes	Read-Only	No	Map TCP/UDP ports to owning PIDs and correlate with `WEBAPP_PORTS.md`.
`win_weeder_scan`	`weeder-scan`	weeder	Read-Only	No	Detect orphaned cargo test binaries, headless engines, and runaway workers.
`win_weeder_reap`	`weeder-reap`	weeder	Mutating	**Yes**	Terminate a non-protected process by PID (requires `confirm=true`).
`win_lock_inspect`	`lock-inspect`	fileops	Read-Only	No	Query Win32 Restart Manager (`rstrtmgr.dll`) for processes locking a file/folder.
`win_fs_list`	`fs-list`	fileops	Read-Only	No	List directory entries with configurable recursion depth.
`win_fs_du`	`fs-du`	fileops	Read-Only	No	Recursive directory disk usage breakdown and top file extensions.

MCP Tool	CLI Verb	Domain	Mode	Confirm	Capability
`win_fs_large`	`fs-large`	fileops	Read-Only	No	Find largest files exceeding a byte threshold.
`win_fs_dups`	`fs-dups`	fileops	Read-Only	No	3-stage streaming SHA-256 duplicate file detector.
`win_fs_hash`	`fs-hash`	fileops	Read-Only	No	Compute streaming FIPS 180-4 SHA-256 digest of a file.
`win_fs_grep`	`fs-grep`	fileops	Read-Only	No	Search text files for a substring while skipping binaries and build dirs.
`win_fs_backup_write`	`fs-backup-write`	fileops	Mutating	**Yes**	Write UTF-8 file after creating a mandatory timestamped `.bak` backup.
`win_fs_undo_backup`	`fs-undo-backup`	fileops	Mutating	**Yes**	Restore the most recent `.bak` backup for a file.
`win_fs_archive_inspect`	`fs-archive-inspect`	fileops	Read-Only	No	Inspect entries and compression ratios inside a `.zip` archive.
`win_fs_archive_create`	`fs-archive-create`	fileops	Mutating	**Yes**	Create a deflated `.zip` archive from a file or directory.
`win_svc_list`	`svc-list`	sysadmin	Read-Only	No	List Windows services with state, display name, and PID.
`win_svc_control`	`svc-control`	sysadmin	Mutating	**Yes**	Start, stop, or restart a Windows service via `sc.exe` (NSSM-safe).
`win_startup_list`	`startup-list`	sysadmin	Read-Only	No	List startup autorun entries across `HKCU` / `HKLM` Run keys and Startup folders.
`win_task_list`	`task-list`	sysadmin	Read-Only	No	List Windows Scheduled Tasks via `schtasks`.
`win_reg_read`	`reg-read`	sysadmin	Read-Only	No	Read values and subkeys from a Windows Registry key.
`win_reg_write`	`reg-write`	sysadmin	Mutating	**Yes**	Write an `HKCU` Registry value with automatic `.reg` backup export.
`win_event_tail`	`event-tail`	sysadmin	Read-Only	No	Tail recent Warning/Error/Critical events from Windows Event Logs.

5. Defense-in-Depth Safety Invariants

Five Non-Bypassable Guards Enforced in `wincraft-core`:

1. **Explicit Confirmation Gate:** Every mutating verb (`weeder_reap`, `fs_backup_write`, `fs_undo_backup`, `fs_archive_create`, `svc_control`, `reg_write`) refuses execution unless `confirm: true` is explicitly supplied.
2. **Global Read-Only Lock:** Passing `--read-only`, setting `WINCRAFT_READ_ONLY=1`, or toggling the UI Read-Only Guard blocks all mutating commands across CLI, MCP, and GUI.
3. **Mandatory Pre-Write Backups:** `fs_backup_write` creates a timestamped `.bak` sibling before modifying any existing file; `reg_write` exports a timestamped `.reg` snapshot to `%LOCALAPPDATA%\wincraft\backups\` before touching `HKCU`.
4. **Protected Process Allowlist:** Critical Windows and developer host processes (`csrss.exe`, `lsass.exe`, `svchost.exe`, `explorer.exe`, `nssm.exe`, `pwsh.exe`, `code.exe`, `claude.exe`) can never be reaped.
5. **NSSM Service Discipline:** `svc_control` inspects `sc.exe qc` for NSSM wrappers and strictly uses `sc.exe stop` / `sc.exe start` rather than killing child processes.